# Dr. Shivam Singh (Ph.D.)

*Postdoctoral Research Associate, University of Virginia, VA, USA*

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## Career goal

My career goal is to contribute to next-generation climate science by improving how extreme hydro-meteorological events are understood, modeled, and translated into actionable information for society. I aim to build research that bridges climate dynamics, high-resolution environmental modeling, and modern artificial intelligence, with an emphasis on developing reliable methods that generalize across regions, climate regimes, and evolving risk landscapes. In the long term, I aspire to become an independent researcher and educator, leading interdisciplinary collaborations and mentoring students to address urgent challenges in climate risk and resilience.

## Academic records

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|---|---|
| **Ph.D.**<br>7.25/10 | **Civil Engineering** — **Indian Institute of Technology Indore, M.P., India**<br>July 2019 - November 2023 |
| **M.Tech**<br>8.03/10 | **Civil Engineering** — **Maulana Azad National Institute of Technology, M.P., India**<br>July 2017 - June 2019 |
| **B.Tech**<br>7.82/10 | **Civil Engineering** — **Maulana Azad National Institute of Technology, M.P., India**<br>July 2013 - June 2017 |
| **12$^{th}$ class**<br>86.00% | **Science** — **Chitrakoot Inter College, Chitrakoot, Uttar Pradesh, India**<br>June, 2011 |
| **10$^{th}$ class**<br>79.00% | **Science** — **G.D.N.D.S.N. Inter College, Chitrakoot, Uttar Pradesh, India**<br>June, 2009 |

## Experiences

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|---|---|
| August 01, 2024<br>-Present | Advanced AI for Simultaneous Bias-Correction and Down-scaling of Climate Model Projections<br>*Postdoctoral Research Associate* Environmental Institute, University of Virginia, USA |
| July 24, 2023<br>-July 14, 2024 | Technological Innovation and Intellectual Property<br>*Project Associate I*, Department of Civil Engineering, IIT Indore |

## Publications

**18 peer-reviewed articles | 500+ citations | h-index: 13**

1. **Singh, S.**, Papalexiou, S. M., Abdelmoaty, M. H., Hartvigsen, T., & Mamalakis, A. (2026). Comprehensive intercomparison of generative AI models for super-resolution precipitation downscaling across hydroclimatic regimes. *Geoscientific Model Development* (Under Review).
2. **Singh, S.**, Papalexiou, S. M., Abdelmoaty, M. H., Hartvigsen, T., & Mamalakis, A. (2025). Correcting dry/wet classification bias in precipitation downscaling via generative adversarial networks. *Environmental Data Science* (Revision Submitted).
3. Abdelmoaty, M. H., Papalexiou, S. M., Mamalakis, A., **Singh, S.**, Coia, V., Hairabedian, M., Szeftel, P., & Grover, P. (2025). Does non-stationarity affect GAN-based downscaling? Insights from high-resolution WRF simulations over the Canadian Prairies. *Artificial Intelligence for the Earth Systems* (Revision Submitted).
4. Abdelmoaty, M. H., Papalexiou, S. M., Mamalakis, A., **Singh, S.**, Coia, V., Hairabedian, M., Szeftel, P., & Grover, P. (2025). Generative adversarial networks for downscaling hourly precipitation in the Canadian Prairies. *Journal of Geophysical Research: Machine Learning and Computation*, 2, e2025JH000678.
5. **Singh, S.**, Yadav, A., & Goyal, M. K. (2024). Univariate and bivariate spatiotemporal characteristics of heat waves, and relative influence of large-scale climate oscillations over India. *Journal of Hydrology*, 628, 130596.
6. **Singh, S.**, Goyal, M. K., & Saikumar, E. (2024). Assessing climate vulnerability of Ramsar wetlands through CMIP6 projections. *Water Resources Management*, 38, 647–669.
7. Rautela, K. S., **Singh, S.**, & Goyal, M. K. (2024). Resilience to air pollution: A novel approach for detecting and predicting aerosol atmospheric rivers within Earth system boundaries. *Earth Systems and Environment*, 9, 93–115.

8. Rautela, K. S., **Singh, S.**, & Goyal, M. K. (2024). Aerosol atmospheric rivers: Patterns, impacts, and societal insights. *Environmental Science and Pollution Research*, 31, 50537–50559.
9. Rautela, K. S., **Singh, S.**, & Goyal, M. K. (2024). Characterizing the spatiotemporal distribution, detection, and prediction of aerosol atmospheric rivers on a global scale. *Journal of Environmental Management*, 351, 119675.
10. **Singh, S.**, & Goyal, M. K. (2023). An innovative approach to predict atmospheric rivers: Exploring convolutional autoencoder. *Atmospheric Research*, 289, 106754.
11. **Singh, S.**, Goyal, M. K., & Jha, S. (2023). Role of large-scale climate oscillations in precipitation extremes associated with atmospheric rivers: Nonstationary framework. *Hydrological Sciences Journal*, 68(3), 395–411.
12. **Singh, S.**, & Goyal, M. K. (2023). Enhancing climate resilience in businesses: The role of artificial intelligence. *Journal of Cleaner Production*, 418, 138228.
13. **Singh, S.**, Kumar, N., Goyal, M. K., & Jha, S. (2023). Relative influence of ENSO, IOD, and AMO over spatiotemporal variability of hydroclimatic extremes in Narmada basin, India. *AQUA – Water Infrastructure, Ecosystems and Society*, 72(4), 520–539.
14. Goyal, M. K., **Singh, S.**, & Jain, V. (2023). Heat waves characteristics intensification across Indian smart cities. *Scientific Reports*, 13, 14786.
15. Kumar, N., Patel, P., **Singh, S.**, & Goyal, M. K. (2023). Understanding non-stationarity of hydroclimatic extremes and resilience in Peninsular catchments, India. *Scientific Reports*, 13, 12524.
16. Meghani, S., **Singh, S.**, Kumar, N., & Goyal, M. K. (2023). Predicting the spatiotemporal characteristics of atmospheric rivers: A novel data-driven approach. *Global and Planetary Change*, 231, 104295.
17. Gupta, V., Rakkasagi, S., Rajpoot, S., El Immanni, H. S., & **Singh, S.** (2023). Spatiotemporal analysis of Imja Lake to estimate the downstream flood hazard using the SHIVEK approach. *Acta Geophysica*, 71, 2233–2244.
18. **Singh, S.**, Prasad, B., & Tiwari, H. L. (2021). Sedimentation analysis for a reservoir using remote sensing and GIS techniques. *ISH Journal of Hydraulic Engineering*, 29(1), 71–79.

## Patent(s)

1. Goyal, M. K. & **Singh, S.**. (2023). Method and system for determining spatiotemporal characteristics and predicting the state of Atmospheric Rivers (AR). Filed in the Indian Patent Office on February 15, 2024. Application No. 202421010646.

## Book(s)

1. Goyal, M. K. & **Singh, S.**. Understanding Atmospheric Rivers Using Machine Learning. *SpringerBriefs in Applied Sciences and Technology* ISBN: 978-3-031-63477-2. Springer.

## Book chapters

1. Vivekanandan, N., **Singh, S.**, & Goyal, M.K. (2023). Comparison of Probability Distributions for Extreme Value Analysis and Predicting Monthly Rainfall Pattern Using Bayesian Regularized ANN. In *Ecosystem Restoration: Towards Sustainability and Resilient Development.* (pp. 271-294). Springer.
2. **Singh, S.**, Dubey, S., Kumar, N., Goyal, M. K., & Pal, I. (2020). Psychological Impacts of COVID-19. Integrated Risk of Pandemic: Covid-19 Impacts, Resilience and Recommendations, 153–168.

## Conference Presentations

1. **American Geophysical Union Fall Meeting 2025**, New Orleans, LA, USA.
   **Singh, S.**, Abdelmoaty, M. H., Papalexiou, S. M., Hartvigsen, T., & Mamalakis, A. *Generative models for super-resolution precipitation downscaling across U.S. hydroclimatic regimes.* Poster presentation, December 15–19, 2025.
2. **International Conference on Smart Systems and Advanced Computing (Syscom 2022)**, Macau, China.
   **Singh, S.** & Goyal, M. K. *Predicting atmospheric rivers employing a deep learning algorithm: Autoencoder.* Oral presentation (virtual), December 30–31, 2022.
3. **National Symposium on Tropical Meteorology**, IISER Bhopal, Madhya Pradesh, India.
   **Singh, S.** & Goyal, M. K. *Heat waves in India: Past and future perspectives.* Oral presentation, November 29–December 2, 2022.
4. **36$^{th}$ Madhya Pradesh Young Scientist Congress**, Madhya Pradesh, India.
   **Singh, S.** & Goyal, M. K. *Role of large-scale climate oscillations over hydroclimatic extremes of Narmada Basin.* Oral presentation, March 23–26, 2021.
5. **26$^{th}$ International Conference on Hydraulics, Water Resources, Environmental and Coastal Engineering (HYDRO 2021)**, NIT Rourkela, India.
   **Singh, S.** & Goyal, M. K. *Heat waves in India: Past and future perspectives.* Oral presentation, March 2021.

## Honors and Awards

- **Ramanujan Fellowship (2025)** awarded by the Anusandhan National Research Foundation (ANRF), Government of India.
- **Young Scientist Award** awarded by the Madhya Pradesh Council of Science and Technology, Bhopal (36$^{th}$ Madhya Pradesh Youth Science Congress, 2021).
- **Second Place – Research Showcase Competition (2025)** at the Postdoctoral Research Symposium, University of Virginia, USA.
- **Best Research Synopsis Award** (PhD Colloquium 2022) jointly organized by Atal Bihari Institute for Good Governance and Policy Analysis (AIGGPA), Bhopal and Indian Council of Social Science Research (ICSSR), New Delhi.
- **Best Poster Presentation Award** at Research and Industrial Conclave (2022), Indian Institute of Technology Indore.
- **Prime Minister Scholarship** awarded by the Government of India for academic merit during bachelor degree.

## Research interests

- Hydro-meteorological Extremes
- Climate Variability and Change
- Atmospheric Rivers
- Climate Downscaling and Bias Correction
- Artificial Intelligence for Earth System Science

## Responsibilities

- Vice-Chair of UVA Postdoc Advocacy Committee
- Teaching Assistant, IIT Indore, 2019-23
- Coordinated Training Programs, IIT Indore, 2022-23
- INSPIRE Club head, NIT Bhopal, 2016-17
- Volunteer as Judge for Departmental Symposiums

## Proficiency

**Technical Expertise**

- Climate Data Handling
- Hydrological Modeling
- Machine Learning

**Analytical Techniques**

- Statistical Analysis
- Probabilistic Analysis
- Risk Mapping

**Software Packages**

- Python, Pytorch, Tensorflow
- Microsoft Office
- ArcGIS, SWAT

## Languages

- Hindi ★★★★★
- English ★★★★★
- Sanskrit ★★☆☆☆

## Hobbies

- Reading Ancient Vedic Texts
- Playing and Watching Cricket
- Memorizing Sanskrit Verses

## References*

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*Hereby, I declare that all the information furnished above is true.*

Date : March 5, 2026
Place : Charlottesville, Virginia

**Shivam Singh**